

Worksheet 2.1

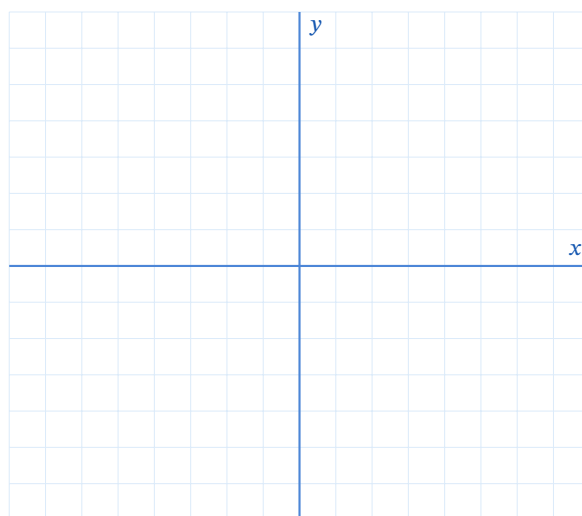
Name: _____

1. Review bridge: a shifted quadratic.

Let

$$g(x) = (x - 2)^2 + 3.$$

- (a) Starting from $y = x^2$, state the horizontal and vertical shifts.
- (b) State the vertex and the axis of symmetry.
- (c) Graph g using the vertex and at least two additional points.



Calculations:



2. Constant rate of change.

The values of a function f are shown below.

x	-1	0	1	2
$f(x)$	2	5	8	11

- Determine whether f is linear. Support your answer using rates of change.
- If f is linear, find its slope.
- Find a formula for $f(x)$.
- State the y -intercept as an ordered pair.

